Dantrolene: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Dantrolene: Patient drug information" and "Dantrolene: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Hepatotoxicity (capsule):

Dantrolene has a potential for hepatotoxicity; do not use in conditions other than those recommended. Symptomatic hepatitis (fatal and nonfatal) has been reported at various dose levels of the drug. The incidence reported in patients taking dosages of up to 400 mg/day is much lower than in those taking dosages of 800 mg or more per day. Even sporadic short courses of these higher dose levels within a treatment regimen markedly increase the risk of serious hepatic injury. Liver dysfunction as evidenced by blood chemical abnormalities alone (liver enzyme elevations) has been observed in patients exposed to dantrolene for varying periods of time. Overt hepatitis has occurred at varying intervals after initiation of therapy, but has been most frequently observed between the third and twelfth month of therapy. The risk of hepatic injury appears to be greater in females, patients >35 years of age, and patients taking other medication(s). Spontaneous reports suggest a higher proportion of hepatic events with fatal outcome in elderly patients receiving dantrolene; however, the majority of these cases were complicated with confounding factors such as concurrent illnesses and/or concomitant potentially hepatotoxic medications. Use dantrolene only in conjunction with appropriate monitoring of hepatic function, including frequent determination of AST or ALT. If no observable benefit is derived from dantrolene after a total of 45 days, discontinue therapy. Prescribe the lowest possible effective dose for the individual patient.

Brand Names: US

- Dantrium;
- Revonto;
- Ryanodex

Brand Names: Canada

- Dantrium

Pharmacologic Category

- Skeletal Muscle Relaxant

Dosing: Adult

Malignant hyperthermia, preoperative prophylaxis

Malignant hyperthermia, preoperative prophylaxis:

Note: Although included as an FDA-approved use in the manufacturer's prescribing information for preoperative prevention of malignant hyperthermia in high-risk patients, prophylaxis is generally not recommended in clinical practice. The use of nontriggering agents and the use of charcoal/carbon filter between the breathing circuit and the anesthesia machine has reduced the risk of malignant hyperthermia in susceptible patients (Ref).

IV: 2.5 mg/kg; administer ~75 minutes prior to anesthesia; repeat doses as needed based on clinical response and duration of surgery.

Oral: 4 to 8 mg/kg/day in 3 to 4 divided doses, begin 1 to 2 days prior to surgery with last dose ~3 to 4 hours before scheduled surgery.

Malignant hyperthermia, treatment

Malignant hyperthermia, treatment:

Note: Administer dantrolene as soon as possible after the identification of clinical symptoms/crisis. Discontinue all malignant hyperthermia–triggering agents (eg, volatile anesthetics gases, succinylcholine) once hyperthermia crisis is recognized and administer supportive care. General treatment goals: end tidal CO_2 <45 mm Hg, normal minute ventilation, core temperature <38ºC (100.4ºF), heart rate stable and decreasing, muscular rigidity resolved (Ref).

Malignant hyperthermia crisis: IV: Initial: 2.5 mg/kg; monitor patient continuously and administer repeat doses of 2.5 mg/kg every 5 minutes until symptoms subside and treatment goals are reached. Most patients will respond to a cumulative dose of ≤10 mg/kg; however, >10 mg/kg may be necessary in patients with persistent contractures or rigidity (Ref).

Recurrence of malignant hyperthermia after initial treatment: Note: Monitor for recurrence (eg, increased muscular rigidity, inappropriate hypercarbia, unexplained metabolic acidosis, inappropriate temperature increase) in PACU/ICU for at least 24 hours (Ref).

IV: 1 mg/kg every 4 to 6 hours or 0.25 mg/**kg**/hour continuous infusion for 24 hours or longer if clinically indicated; stop therapy or change dosing interval to every 8 to 12 hours when all of the following are met: metabolic stability for 24 hours, core temperature <38ºC (100.4ºF), decrease CK, myoglobinuria absent, no muscle rigidity (Ref).

MHAUS 24-hour MH Hotline (for emergencies only):

United States: 1-800-644-9737.

Outside the United States: 001-209-417-3722.

Neuroleptic malignant syndrome, moderate to severe

Neuroleptic malignant syndrome, moderate to severe (off-label use):

Note: Consider for use in combination with supportive care, benzodiazepines, and/or dopaminergic agents (eg, bromocriptine) in patients with severe symptoms at presentation (eg, hyperthermia, evidence of rhabdomyolysis) and for those not responding to initial withdrawal of medication and supportive care (Ref).

IV: Initial: 1 to 2.5 mg/kg initially; if rapid resolution of hyperthermia and rigidity is observed, may follow with 1 mg/kg every 6 hours (maximum cumulative daily dose: 10 mg/kg/day) (Ref). After the patient is stabilized and symptoms have resolved, consider taper over days to weeks rather than abrupt discontinuation (Ref).

Spasticity, chronic

Spasticity, chronic: Oral:

Note: Dose should be titrated and individualized for maximum effect; use the lowest dose compatible with optimal response. Some patients may not respond until a higher daily dosage is achieved;

each dose level should be maintained for 7 days to determine patient response. If no further benefit observed with the higher dose level, then decrease dosage to previous dose level. Because of the potential for hepatotoxicity, stop therapy if benefits are not evident within 45 days.

Initial: 25 mg once daily for 7 days; increase to 25 mg 3 times daily for 7 days, increase to 50 mg 3 times daily for 7 days, and then increase to 100 mg 3 times daily; some patients may require 100 mg 4 times daily; maximum dose: 400 mg/day

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, need for dosage adjustment unlikely as limited excretion in the urine.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; use of oral dantrolene in patients with active liver disease (eg, hepatitis and cirrhosis) is contraindicated.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Dantrolene: Pediatric drug information")

Malignant hyperthermia, treatment

Malignant hyperthermia, treatment:

Note: Utilize actual body weight (Ref) for dosing calculations. There is no recommended maximum total dose. Although FDA approved, routine dantrolene for the prevention of malignant hyperthermia is not recommended (Ref).

If questions, contacting the MH Hotline (see below) may be helpful.

24-hour MH Hotline (for emergencies only):

United States: 1-800-644-9737

Outside the United States: 001-209-417-3722

Crisis: Infants, Children, and Adolescents: IV: Initial: 2.5 mg/kg, then 1 or 2.5 mg/kg every 5 minutes as frequently as needed until treatment goals reached (Ref). Doses >10 mg/kg may be required for patients with persistent contractures or rigidity; however, if no resolution, reassess diagnosis (Ref).

Postcrisis follow-up:

IV: Infants, Children, and Adolescents: IV: 1 mg/kg/dose every 4 to 6 hours (preferred to minimize extravasation risk) if within 6 hours of initial reaction; if outside of 6-hour window, a higher dose of 2 or 3 mg/kg/dose may be needed. Treatment may be stopped or interval between doses increased to every 8 to 12 hours when the following criteria are met: metabolic stability for 24 hours, core temperature <38°C, creatinine kinase continues to decrease, no evidence of ongoing myoglobinuria, and muscle rigidity has resolved (Ref).

Oral: **Note:** Although FDA approved for this indication, expert guidelines do not recommend use (Ref): Infants, Children, and Adolescents: Oral: 4 to 8 mg/kg/**day** in 4 divided doses for 1 to 3 days.

Spasticity, chronic

Spasticity, chronic: Note: Dosing should be individualized based on patient response and tolerability. Titrate to desired effect; if no further benefit is observed at a higher dosage, decrease dose to previous lower dose. Use of the lowest dose associated with the optimal response is recommended:

Children ≥5 years and Adolescents: Oral:

<50 kg: Initial: 0.5 mg/kg/dose once or twice daily for 7 days; increase to 0.5 mg/kg/dose 3 times daily for 7 days, then increase to 1 mg/kg/dose 3 times daily for 7 days, and then increase to 2 mg/kg/dose 3 times daily; some patients may require a dose 4 times daily; maximum daily dose: 12 mg/kg/**day** up to 400 mg/**day** (Ref).

≥50 kg: Initial: 25 mg once daily for 7 days; increase to 25 mg 3 times daily for 7 days, then increase to 50 mg 3 times daily for 7 days, and then increase to 100 mg 3 times daily; some patients may require a dose 4 times daily; maximum daily dose: 400 mg/**day**.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents: There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

Infants, Children, and Adolescents: There are no dosage adjustments provided in the manufacturer's labeling; use of oral dantrolene in patients with active liver disease (hepatitis or cirrhosis) is contraindicated.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Postmarketing:

Cardiovascular: Atrioventricular block, flushing, heart failure, phlebitis (including superficial thrombophlebitis) (Brandom 2011), tachycardia, variable blood pressure

Dermatologic: Acneiform eruption (Mowbray 2009), diaphoresis, eczematous rash, hair disease (abnormal growth), pruritus, urticaria

Endocrine & metabolic: Hyperkalemia (Brandom 2011)

Gastrointestinal: Abdominal cramps, anorexia, constipation, diarrhea (including severe diarrhea), dysgeusia, dysphagia, gastric irritation (Brandom 2011), gastrointestinal hemorrhage, nausea (Brandom 2011), sialorrhea (Brandom 2011), vomiting

Genitourinary: Crystalluria, difficulty in micturition, erectile dysfunction, hematuria, nocturia, urinary frequency, urinary incontinence, urinary retention

Hematologic & oncologic: Anemia, aplastic anemia, leukopenia, malignant lymphoma (lymphocytic) (Wan 1980), thrombocytopenia

Hepatic: Hepatitis (Wilkinson 1979), hepatotoxicity (including hepatic necrosis) (Wilkinson 1979), increased liver enzymes (Wilkinson 1979)

Hypersensitivity: Anaphylaxis

Local: Injection-site reaction (including erythema at injection site, injection-site phlebitis, pain at injection site, swelling at injection site)

Nervous system: Asthenia (Brandom 2011), chills, confusion, depression, dizziness (Brandom 2011), drowsiness, fatigue (Brandom 2011), headache, insomnia, malaise, myasthenia (Brandom 2011), nervousness, seizure, speech disturbance, voice disorder

Neuromuscular & skeletal: Back pain, dystonia, limb pain, myalgia

Ophthalmic: Blurred vision, diplopia, epiphora, visual disturbance (Brandom 2011)

Respiratory: Dyspnea (including a feeling of suffocation) (Brandom 2011), oropharyngeal spasm (Locatelli 2014), pleural effusion (with eosinophilia or pericarditis) (Miller 1984), pulmonary edema (Brandom 2011), respiratory depression, respiratory failure (Brandom 2011), respiratory insufficiency (including respiratory muscle weakness and decreased inspiratory capacity) (Javed 2010)

Miscellaneous: Fever

Contraindications

IV: There are no contraindications listed within the manufacturer's labeling.

Oral: Active hepatic disease (eg, cirrhosis, hepatitis); when spasticity is used to maintain upright posture/balance in locomotion or to obtain/maintain increased function.

Canadian labeling (additional contraindications not in US labeling):

IV: Hypersensitivity to dantrolene or any component of the formulation; compromised pulmonary function (eg, obstructive pulmonary disease).

Oral: Hypersensitivity to dantrolene or any component of the formulation; when spasticity is needed to maintain function; compromised pulmonary function (eg, obstructive pulmonary disease); active hepatic disease (eg, cirrhosis, hepatitis).

Warnings/Precautions

Concerns related to adverse effects:

• CNS depression: May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks that require mental alertness (eg, operating machinery or driving).

• Hepatotoxicity: Oral: Has potential for hepatotoxicity; symptomatic hepatitis (fatal and nonfatal) has been reported. Idiosyncratic and hypersensitivity reactions (sometimes fatal) of the liver have also occurred; monitor hepatic function. Discontinue therapy if symptoms compatible with hepatitis, accompanied with abnormal LFTs or jaundice, occur or benefits are not observed within 45 days. Hepatic function usually reverts to normal upon discontinuation; if reinstitution of therapy is necessary, patients should be hospitalized and the drug initiated in very small and gradual dose increases.

• Muscle weakness: Loss of grip strength, weakness in the legs, dyspnea, respiratory muscle weakness, dysphagia, and decreased inspiratory capacity has occurred with IV dantrolene. Patients should not ambulate without assistance until they have normal strength and balance. Monitor patients for the adequacy of ventilation and for difficulty swallowing/choking.

• Photosensitivity: Oral therapy may cause a photosensitivity reaction; use with caution during exposure to sunlight.

• Pleural effusion: Pleural effusion with associated eosinophilia may occur.

Disease-related concerns:

• Cardiovascular disease: Use oral therapy with caution in patients with severely impaired cardiac function due to myocardial disease.

• Hepatic disease: Use oral therapy with caution in patients with history of hepatic disease or dysfunction; use is contraindicated in patients with active hepatic disease (eg, cirrhosis, hepatitis).

• Respiratory disease: Use oral therapy with caution in patients with impaired pulmonary function (particularly in obstructive pulmonary disease).

Dosage form specific issues:

• Extravasation: Alkaline solution; may cause tissue necrosis if extravasated (vesicant); ensure proper needle or catheter placement prior to and during infusion; avoid extravasation.

• Mannitol: Injection may contain mannitol.

• Polysorbate 80: Some dosage forms may contain polysorbate 80 (also known as Tweens). Hypersensitivity reactions, usually a delayed reaction, have been reported following exposure to pharmaceutical products containing polysorbate 80 in certain individuals (Isaksson 2002; Lucente 2000; Shelley, 1995). Thrombocytopenia, ascites, pulmonary deterioration, and renal and hepatic failure have been reported in premature neonates after receiving parenteral products containing polysorbate 80 (Alade 1986; CDC 1984). See manufacturer's labeling.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Capsule, Oral, as sodium:

Dantrium: 25 mg [contains fd&c yellow #6 (sunset yellow)]

Generic: 25 mg, 50 mg, 100 mg

Solution Reconstituted, Intravenous, as sodium:

Dantrium: 20 mg (1 ea)

Solution Reconstituted, Intravenous, as sodium [preservative free]:

Revonto: 20 mg (1 ea)

Generic: 20 mg (1 ea)

Suspension Reconstituted, Intravenous, as sodium:

Ryanodex: 250 mg (1 ea) [contains polysorbate 80]

Generic Equivalent Available: US

May be product dependent

Pricing: US

Capsules (Dantrium Oral)

25 mg (per each): $1.28

Capsules (Dantrolene Sodium Oral)

25 mg (per each): $0.96 - $1.07

50 mg (per each): $1.56 - $1.73

100 mg (per each): $1.95 - $2.17

Solution (reconstituted) (Dantrium Intravenous)

20 mg (per each): $106.37

Solution (reconstituted) (Dantrolene Sodium Intravenous)

20 mg (per each): $85.09

Solution (reconstituted) (Revonto Intravenous)

20 mg (per each): $84.00

Suspension (reconstituted) (Ryanodex Intravenous)

250 mg (per each): $4,215.86

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Capsule, Oral, as sodium:

Dantrium: 25 mg [contains fd&c yellow #6 (sunset yellow)]

Solution Reconstituted, Intravenous, as sodium:

Dantrium: 20 mg (1 ea)

Generic: 20 mg (1 ea)

Extemporaneous Preparations

A 5 mg/mL oral suspension may be made with dantrolene capsules, a citric acid solution, and either simple syrup or syrup BP (containing 0.15% w/v methylhydroxybenzoate). Add the contents of five 100 mg dantrolene capsules to a citric acid solution (150 mg citric acid powder in 10 mL water); mix while adding the chosen vehicle in incremental proportions to **almost** 100 mL. Transfer to a calibrated bottle and add quantity of vehicle sufficient to make 100 mL. Label ”shake well” and ”refrigerate”. Simple syrup suspension is stable for 2 days refrigerated; syrup BP suspension is stable for 30 days refrigerated (Ref).

Administration: Adult

IV:

Malignant hyperthermia crisis/recurrence: Administer as a rapid continuous IV push, through a large bore IV if possible (do not delay administration if large bore IV is not available) (Ref).

Other indications: Administer over at least 1 minute (Ryanodex) or 1 hour (Dantrium, Revonto) (Ref).

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm compresses; initiate hyaluronidase antidote (if indicated) (Ref).

Hyaluronidase: **Intradermal or SUBQ:** Inject a total of 1 mL (15 units/mL) as 5 separate 0.2 mL injections (using a tuberculin syringe) around the site of extravasation; if infiltrated catheter remains in place, administer through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution (Ref).

Preparation for Administration: Adult

Injection, powder for reconstitution:

Dantrium, Revonto: Reconstitute vial by adding 60 mL of sterile water for injection only (**not bacteriostatic water for injection**); shake well until clear; avoid glass bottles for IV infusion due to potential for precipitate formation.

Ryanodex: Reconstitute vial by adding 5 mL of sterile water for injection only (**not bacteriostatic water for injection**); shake well (suspension is an orange color). Do not dilute or transfer the suspension to another container to infuse the product.

Administration: Pediatric

Oral: Administer liquid formulation (extemporaneously prepared) using an accurate measuring device; do not use household tablespoon.

Parenteral:

Dantrium, Revonto: Administer crisis doses by rapid IV injection, preferably in large-bore IV or central line.

Ryanodex: Administer crisis doses by rapid IV injection; follow-up doses should be administered over ≥1 minute.

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. If extravasation occurs, stop infusion immediately and disconnect (leave cannula/needle in place); gently aspirate extravasated solution (do **NOT** flush the line); remove needle/cannula; elevate extremity.

Preparation for Administration: Pediatric

Parenteral:

Dantrium, Revonto: Reconstitute vial by adding 60 mL of SWFI only **(not bacteriostatic water for injection)**, resultant concentration: 0.333 mg/mL; shake vial for ~20 seconds or until solution is clear; no further dilution required; avoid glass bottles for IV infusion due to potential for precipitate formation. Protect from direct light. Use within 6 hours after reconstitution.

Ryanodex: Reconstitute vial by adding 5 mL of SWFI only **(not bacteriostatic water for injection)**; resultant concentration: 50 mg/mL; shake vial well (suspension is an orange color). Do not dilute or transfer the suspension to another container to infuse the product. Use within 6 hours after reconstitution.

Vesicant/Extravasation Risk

Vesicant; because of the high pH (~9.5 to 10.3), may cause tissue necrosis if extravasated.

Storage/Stability

Capsules: Store at 20°C to 25°C (68°F to 77°F).

Injection, powder for reconstitution: Protect from light. Use reconstituted solution within 6 hours of preparation.

Dantrium: Store unreconstituted vials and reconstituted solutions at 15°C to 30°C (59°F to 86°F).

Revonto: Store unreconstituted vials and reconstituted solutions at 20°C to 25°C (68°F to 77°F).

Ryanodex: Store unreconstituted vials at 20°C to 25°C (68°F to 77°F); excursions are permitted between 15°C and 30°C (59°F and 86°F). Store reconstituted solutions at 20°C to 25°C (68°F to 77°F).

Use: Labeled Indications

Chronic spasticity: Oral: Treatment of spasticity associated with upper motor neuron disorders (eg, spinal cord injury, stroke, cerebral palsy, or multiple sclerosis).

Malignant hyperthermia: Treatment of malignant hyperthermia crisis; following a malignant hyperthermia crisis to prevent recurrence; preoperative prophylaxis of malignant hyperthermia.

Use: Off-Label: Adult

Neuroleptic malignant syndrome, moderate to severe

Medication Safety Issues

Sound-alike/look-alike issues:

Dantrium may be confused with danazol, Daraprim

Revonto may be confused with Revatio

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drugs (pediatric liquid medications requiring measurement) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021).

Metabolism/Transport Effects

Substrate of CYP3A4 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylinoxide: CNS Depressants may increase CNS depressant effects of Amitriptylinoxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buclizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Calcium Channel Blockers: Dantrolene may increase hyperkalemic effects of Calcium Channel Blockers. Dantrolene may increase negative inotropic effects of Calcium Channel Blockers. Management: Consider alternatives to the use of dantrolene with calcium channel blockers during the management of malignant hyperthermia when possible, as product labeling states that the concomitant use of these agents is not recommended. *Risk D: Consider Therapy Modification*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Chlorpheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: CNS Depressants may increase CNS depressant effects of CloZAPine. *Risk C: Monitor*

CNS Depressants: Dantrolene may increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

Dexketoprofen: May increase adverse/toxic effects of Dantrolene. Management: Consider alternatives to the coadministration of dexketoprofen and dantrolene when possible, as product labeling states the concomitant use of these agents is not recommended. If combined, monitor for dantrolene toxicities (eg, drowsiness, fatigue). *Risk D: Consider Therapy Modification*

DexmedeTOMIDine: CNS Depressants may increase CNS depressant effects of DexmedeTOMIDine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

DiphenhydrAMINE (Systemic): CNS depressant effects of CNS Depressants and DiphenhydrAMINE (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Estrogen Derivatives: May increase hepatotoxic effects of Dantrolene. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

FluPHENAZine: CNS Depressants may increase CNS depressant effects of FluPHENAZine. *Risk C: Monitor*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

HydrOXYzine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketamine: CNS Depressants may increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRAcetam: CNS Depressants may increase CNS depressant effects of LevETIRAcetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methoxyflurane: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: CNS Depressants may increase CNS depressant effects of Mirtazapine. *Risk C: Monitor*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Muscle Relaxants (Centrally Acting): Dantrolene may increase adverse neuromuscular effects of Muscle Relaxants (Centrally Acting). Specifically, neuromuscular block may be potentiated. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

OLANZapine: CNS depressant effects of OLANZapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): CNS Depressants may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). Management: Consider alternatives to this combination when possible. If combined, dose reduction or discontinuation of one or more CNS depressants (including the oxybate salt product) should be considered. Interrupt oxybate salt treatment during short-term opioid use *Risk D: Consider Therapy Modification*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Pimozide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propofol: CNS Depressants may increase CNS depressant effects of Propofol. *Risk C: Monitor*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

QUEtiapine: CNS Depressants may increase CNS depressant effects of QUEtiapine. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

RisperiDONE: CNS depressant effects of CNS Depressants and RisperiDONE may increase with coadministration. *Risk C: Monitor*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIRole: CNS Depressants may increase sedative effects of ROPINIRole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Thioridazine: CNS Depressants may increase CNS depressant effects of Thioridazine. *Risk C: Monitor*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

TiaGABine: CNS Depressants may increase CNS depressant effects of TiaGABine. *Risk C: Monitor*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Triprolidine: CNS Depressants may increase CNS depressant effects of Triprolidine. *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: CNS Depressants may increase CNS depressant effects of Valproic Acid and Derivatives. *Risk C: Monitor*

Vecuronium: Dantrolene may increase neuromuscular-blocking effects of Vecuronium. *Risk C: Monitor*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Ziprasidone: CNS Depressants may increase CNS depressant effects of Ziprasidone. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Dantrolene crosses the human placenta.

Cord blood concentrations of dantrolene are similar to those in the maternal plasma at term and dantrolene can be detected in the newborn serum at delivery. Adverse events were not observed in the newborn following maternal doses of 100 mg/day administered orally prior to delivery (Shime 1988).

Uterine atony has been reported following dantrolene injection after delivery; however, this may be due in part to the mannitol contained in the IV preparation (Shin 1995; Weingarten 1987). Untreated malignant hyperthermia (MH) is a medical emergency; treatment should not be withheld due to pregnancy. Prophylactic use of dantrolene is not routinely recommended in pregnant patients susceptible to MH prior to obstetric surgery; if use is needed, close monitoring of the mother and newborn is recommended (Krause 2004; Norman 1995; Urman 2019).

Breastfeeding Considerations

Dantrolene is present in breast milk.

In a case report, the half-life of dantrolene in breast milk was calculated to be 9 hours; the highest milk concentration was 1.2 mcg/mL following a maternal IV dose; however, the maternal serum concentrations were not reported (Fricker 1998).

Due to the potential for adverse reaction in the breastfeeding infant, breastfeeding is not recommended by the manufacturer; breastfeeding should be interrupted, and breast milk pumped and discarded during treatment and for 3 days after dantrolene administration.

Monitoring Parameters

Nausea, vomiting, and LFTs (baseline and at appropriate intervals thereafter) should be monitored for potential hepatotoxicity; IV administration requires cardiac, BP, respiratory, and extravasation monitoring; motor performance should be monitored for spasticity use.

Malignant hyperthermia: During and post-acute phase: Patient should be observed in a PACU/ICU for at least 24 hours since recrudescence (recurrence) may occur; ECG; vital signs (including core and peripheral temperature); electrolytes, ABG, CK, end tidal CO_2 ($EtCO_2$)/capnography (if intubated), O_2 saturation, urine output, urine myoglobin, coagulation studies (Hopkins 2021; MHAUS 2021).

Neuroleptic malignant syndrome: Electrolytes, PT/INR, aPTT, creatine kinase level, urine output, vital signs (including temperature) (Strawn 2007).

Mechanism of Action

Acts directly on skeletal muscle by interfering with release of calcium ion from the sarcoplasmic reticulum; prevents or reduces the increase in myoplasmic calcium ion concentration that activates the acute catabolic processes associated with malignant hyperthermia

Pharmacokinetics (Adult Data Unless Noted)

Absorption: Oral: 70% (Allen 1988)

Distribution: V_d: 36.4 ± 11.7 L

Metabolism: Hepatic; major metabolites are 5-hydroxy dantrolene and an acetylamino metabolite of dantrolene.

Half-life elimination:

Neonates (at birth): ~20 hours (Shime 1988)

Children 2 to 7 years: 10 hours (range: 8.1 to 14.8 hours) (Lerman 1989)

Adults: 4 to 11 hours

Time to peak: IV: 1 minute post-dose (dantrolene); 24 hours post-dose (5-hydroxy dantrolene)

Excretion: Feces (45% to 50%); urine (25% as unchanged drug and metabolites)

Patient Counseling Points

(For additional information see ”Dantrolene: Patient drug information”)

What is this drug used for?

• It is used in certain patients to relax muscles.

• It is used to treat or prevent a health problem called malignant hyperthermia.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Capsules:

• Feeling dizzy, sleepy, tired, or weak

• Diarrhea

Injection:

• Flushing

• Upset stomach

• Change in voice

• Feeling sleepy

• Dizziness

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

• Liver problems like dark urine, tiredness, decreased appetite, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes

• Choking

• Trouble swallowing

• Shortness of breath

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Capsules:

• Chest pain or pressure

Injection:

• Injection site reaction like any redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Dantrium;
- (AR) Argentina: Dantocris | Dantrocris | Dantrolen;
- (AT) Austria: Dantamacrin | Dantrolen;
- (AU) Australia: Dantrium | Revonto;
- (BD) Bangladesh: Danflex | Danlene | Relatro | Relaxo | Trolen;
- (BE) Belgium: Dantrium;
- (BG) Bulgaria: Dantrium;
- (BR) Brazil: Dantrolen;
- (CH) Switzerland: Dantamacrin;
- (CL) Chile: Dantrium | Dantroleno;
- (CO) Colombia: Dantroleno;
- (DE) Germany: Dantamacrin | Dantrium | Dantrolen;
- (DO) Dominican Republic: Dantrolen;
- (EE) Estonia: Dantrium | Dantrolen;
- (EG) Egypt: Betenolmen | Dantrelax | Relatrolene;
- (FR) France: Dantrium | Dantrium intraveineux;
- (GB) United Kingdom: Agilus | Dantrium;
- (GR) Greece: Dantamacrin | Dantrium | Dantrolene;
- (HK) Hong Kong: Dantrium;
- (HU) Hungary: Dantrolen | Dantrolene;
- (IE) Ireland: Dantrium;
- (IL) Israel: Dantrium;
- (IT) Italy: Dantrium;
- (JP) Japan: Dantrium;
- (KR) Korea, Republic of: Anorex | Dantrolen;
- (LT) Lithuania: Dantrolen;
- (LU) Luxembourg: Dantrium;
- (LV) Latvia: Dantamacrin | Dantrium | Dantrolen;
- (MY) Malaysia: Dantrium;
- (NL) Netherlands: Dantrium;
- (NO) Norway: Agilus | Dantrium | Dantrolen | Dantrolen ahn | Dantrolen campus;
- (NZ) New Zealand: Dantrium;
- (PL) Poland: Dantamacrin | Dantrium | Dantrolen;
- (PR) Puerto Rico: Dantrium;
- (PT) Portugal: Dantrium;
- (QA) Qatar: Dantrium | Revonto;
- (RO) Romania: Dantrium iv;
- (RU) Russian Federation: Dantrium;
- (SI) Slovenia: Dantrium | Dantrolen;
- (SK) Slovakia: Dantrolen;
- (TN) Tunisia: Dantrium;

- (TW) Taiwan: Dantrium | Dantrolene;
- (UY) Uruguay: Dantrolene;
- (ZA) South Africa: Dantrium

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